
Common Entity (Reference)

```
public class Student
{
    public int Id { get; set; }
    public string Name { get; set; }
    public int Age { get; set; }
    public string Department { get; set; }
    public int Marks { get; set; }
}

public DbSet<Student> Students { get; set; }
```

1. Get All Records

```
var students = _context.Students.ToList();
```

2. Get Student By Id

```
var student = _context.Students
    .FirstOrDefault(s => s.Id == 1);
```

3. Get Students with Condition (WHERE)

```
var students = _context.Students
    .Where(s => s.Age > 18)
    .ToList();
```

4. Multiple Conditions

```
var students = _context.Students
```

```
.Where(s => s.Age > 18 && s.Marks > 70)
.ToList();
```

5. Select Specific Columns (Projection)

```
var students = _context.Students
    .Select(s => new
    {
        s.Name,
        s.Marks
    })
.ToList();
```

6. Sorting (ORDER BY)

Ascending

```
var students = _context.Students
    .OrderBy(s => s.Marks)
.ToList();
```

Descending

```
var students = _context.Students
    .OrderByDescending(s => s.Marks)
.ToList();
```

7. Top N Records (LIMIT / TAKE)

```
var topStudents = _context.Students
    .OrderByDescending(s => s.Marks)
    .Take(3)
```

```
.ToList();
```

8. Paging (Skip + Take)

```
var students = _context.Students
    .OrderBy(s => s.Id)
    .Skip(10)
    .Take(5)
    .ToList();
```

9. Check Data Exists (ANY)

```
bool exists = _context.Students
    .Any(s => s.Marks > 90);
```

10. Count Records

```
int count = _context.Students.Count();
```

11. Aggregate Functions

Average

```
var avgMarks = _context.Students
    .Average(s => s.Marks);
```

Max

```
var maxMarks = _context.Students
    .Max(s => s.Marks);
```

Min

```
var minMarks = _context.Students
    .Min(s => s.Marks);
```

12. Group By

```
var grouped = _context.Students
    .GroupBy(s => s.Department)
    .Select(g => new
    {
        Department = g.Key,
        Count = g.Count()
    })
    .ToList();
```

13. Distinct Values

```
var departments = _context.Students
    .Select(s => s.Department)
    .Distinct()
    .ToList();
```

14. Include (Eager Loading - Related Data)

 Example: Student → Department Navigation Property

```
var students = _context.Students
    .Include(s => s.DepartmentNavigation)
    .ToList();
```

15. Find by Primary Key (Fast Lookup)

```
var student = _context.Students.Find(1);
```
